Amkor: qualified packaging capacity must become profitable output

Amkor can earn more from each generation of computing hardware as chips require more complex packaging, testing, power delivery and thermal control. Its customer relationships and expanding US footprint create a route to higher-value work. The shareholder opportunity depends on qualifying those lines, filling them and retaining a profit after materials, engineering and capital costs.

The important new development is Arizona Phase 2. Customer commitments exceeded the first phase's planned space, prompting a much larger campus. Production from the new phase is expected from 2030. Program spending arrives earlier. The active Q4 2027 ceiling is about $79 per share under the retained 20x forward operating-profit method.

Research snapshot	Current assessment
Date / latest actual	21 October 2026. Latest reported quarter: Q2 ended 30 June, released 27 July. The next earnings call is scheduled for 26 October and is outside this cutoff.
Business / exposure	NASDAQ: AMKR. Outsourced semiconductor assembly and test; one accounting segment with several end-market allocations. Covered status does not assert a position.
Analytical score	43 / 60, rounded to 72%. The category review is shown in full near the end.
Protected Excel Base	72; kept separate from analytical judgment.
Valuation basis	20x forward-four-quarter GAAP operating profit. The selected Q4 2027 window is Q4 2027 through Q3 2028.
Valued operating profit	$984 million in that selected four-quarter window.
Share denominator	249,978,000 diluted weighted-average shares, held flat under the existing convention.
Principal uncertainty	Qualification, customer acceptance, utilization, manufacturing margin, new program costs and financing. Product margins and the early new-campus ramp are LOW-confidence estimates.

Money is USD millions unless stated otherwise. Forecasts are model estimates. Human-facing values are rounded to whole numbers; exact source decimals and full model precision are retained. Independently rounded components can differ from a displayed total.

The alpha: regional qualification and more packaging content

Advanced processors increasingly depend on chiplets, dense interconnects, memory integration and demanding test. A qualified packaging partner can capture a larger service wallet when the design requires more complex assembly and verification. Regional supply-chain requirements can make qualified US capacity more valuable to customers that need an end-to-end domestic route.

The market can overfocus on the headline campus budget while missing the sequence between customer engagement, qualification, accepted packages and manufacturing profit. The economic edge comes from faster conversion into well-utilized, higher-value lines. Empty floor space and expensive tools do not establish saleable throughput.

Transmission into earnings	Evidence and qualification
More work per chip / system	May Investor Day describes chiplet architectures and rising advanced-packaging content. Its industry estimate of about $500 million of OSAT opportunity per installed gigawatt is a broad market proxy; no direct Amkor revenue is calculated from it. [A18B, slide 9]
Qualified regional capacity	The September issuer release says customer commitments exceeded the planned first-phase area. This supports expansion; exact commercial terms and accepted package volumes remain undisclosed. [A10]
Higher utilization and mix	Management attributed most Q2 gross-margin expansion to utilization and the remainder to mix. Operating leverage needs funded, accepted production. [A23]
Profit capture	Materials, labor, depreciation, yield losses, support and program costs are deducted. Gross margin, rather than the campus budget, links receipts to operating earnings.

Competing OSATs, foundries and customer in-house packaging remain credible alternatives. Amkor's investment case therefore rests on execution, qualification and geography, with no permanent monopoly assumed.

AMKR / 21 OCTOBER 2026 / FRAMEWORK 8.0

Products and the physical bottleneck

Amkor supplies advanced packaging and test across computing, communications, automotive/industrial and consumer electronics. High-density fan-out, flip-chip and interposer-based integration can connect processors, memory and networking functions. Larger packages, more process steps and thermal constraints can reduce good-unit throughput even when nominal space expands.

TECHNOLOGY SOLUTIONS

SWIFT® Technology

Silicon Wafer Integrated Fan-out Technology (SWIFT) is designed to provide increased I/O and circuit density in a reduced footprint and profile for single and multi-die applications. SWIFT technology enables creation of advanced 3D structures, addressing the need for increased IC integration in emerging mobile and networking applications.

SWIFT® PACKAGING CONSIDERATIONS

Amkor's state-of-the-art High-Density Fan-Out (HDFO) structure called Silicon Wafer Integrated Fan-out Technology (SWIFT) packaging, bridges the gap between TSV and laminate substrates.

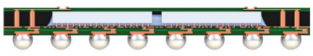
FEATURES

- Polymer dielectrics
- Multi-die and large die capability
- Large body package capability
- Interconnect density down to 2/2 µm
- Cu pillar die interconnect down to 30 µm pitch
- Meets JEDEC MSL2a and MSL3 CLR and BLR requirements

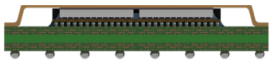
The distinctive characteristics of SWIFT technology are due, in part, to the fine feature capabilities associated with this innovative wafer level packaging technique. This allows more aggressive design rules to be applied compared to traditional WLFO and laminate-based assemblies.

SWIFT Structures and Attributes

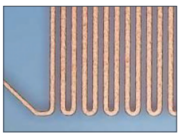
The figure below shows cross-section illustrations of a single die SWIFT and dual-die 3D/Package-on-Package (PoP) SWIFT structures. Although the package appears to be of a typical fine-pitch flip chip construction, it incorporates some unique features not associated with conventional IC packages.



Fan-in PoP SWIFT®



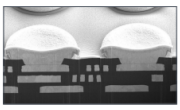
SWIFT® on Substrate



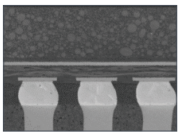
Plated Cu RDL L/S down to 2 µm



Multilayer Redistribution



Multilayer RDL Cross Section



Tall Cu Pillar for PoP Applications

Enabling Technologies for SWIFT Packaging

Key assembly technologies enable the creation of these distinctive SWIFT packaging features and attributes. Through the use of stepper photo imaging equipment, 2/2 µm line/space features can be achieved, enabling very high density die-to-die connections required for SoC partitioning and networking applications where 2.5D TSV would typically be used. Fine-pitch die micro bumps provide a high-density interconnect for advanced products, such as application processors and baseband devices. In addition, tall Cu pillars enable a high-density vertical interface for mounting advanced memory devices on the top of the SWIFT structure.

Amkor Technology®

A19: original Amkor SWIFT technical data sheet, September 2018, page 1. Architecture and historical platform attributes explain the interconnect mechanism. These specifications are not current yield, market share, package pricing or 2026 capacity guidance.

Q2 advanced-product revenue was about $1,557 million, covering flip chip, memory, wafer processing and related test. That classification is broader than AI or high-density fan-out. It cannot all be counted as a new AI sales pool. [A02P, physical pages 12-13]

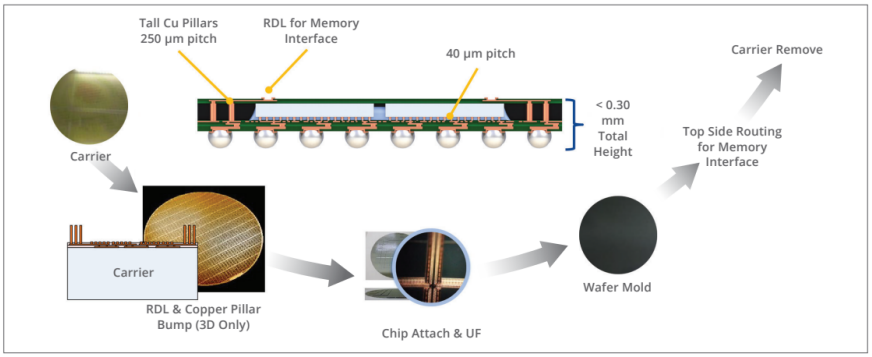
SWIFT® Technology

SWIFT Packaging Process Flow

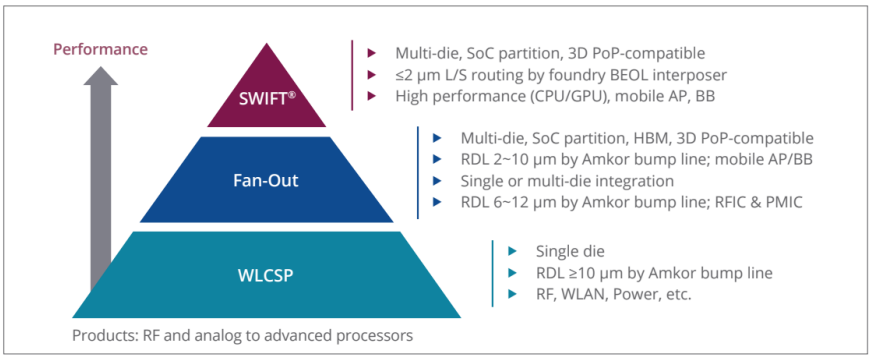
SWIFT product attributes are realized by applying a unique process flow that incorporates both flip chip assembly and wafer level processing techniques, as shown below.

Reliability Data					
Component Level				Board Level	
MSL2aA, MSLA3 Precon	Unbiased HAST 192 hrs	TCB 1500 cycles	HTS 1000 hrs	Drop	Temp Cycle
Pass	Pass	Pass	Pass	Pass	Pass

SWIFT Packaging Technology Process Flow



Advanced Wafer Product Positioning



Visit amkor.com or email sales@amkor.com for more information.

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A19: original Amkor process flow, September 2018, page 2. It shows the manufacturing sequence and integration options. Original labels and qualifications are retained; no current production economics are inferred from the historical diagram.

Why performance, thermal control and yield matter

The constrained resource is qualified good-package output. A denser interconnect is useful only if the package can be built reliably, supplied with power, cooled and tested at the required yield. Process complexity can raise the customer's willingness to pay, while simultaneously raising Amkor's material, tooling and engineering costs.

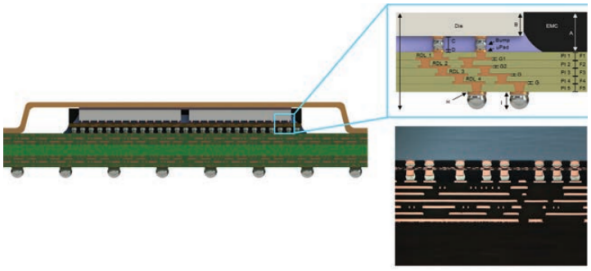


Figure 3. S-SWIFT® packaging employs multi-layer copper and organic dielectric RDLs.

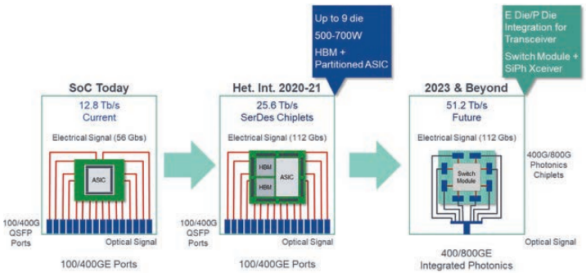


Figure 4. The progression of the switch roadmap will lead to integrated silicon photonics solutions.

bandwidth memory (HBM) provides the complete solution. The number of HBMs in these designs has been increasing with some solutions looking into as many as 6-8 HBM stacks in one package. Discrete I/O chips are used because SerDes I/O chips do not scale well with process nodes and may be able to stay in an older, less expensive silicon node. This allows the reuse of intellectual property (IP) as this same die can be used in other products and increases the total number of I/O's beyond what is possible in a single SOC die.

There are many ways to address package-level integration of different types of semiconductor technologies or Heterogeneous Integration. One is the multi-chip module (MCM) approach with different die attached to the package substrate. A second approach uses high-density modules, examples are 2.5D construction, which uses a silicon interposer to connect the different dies together. Another approach uses high-density fan-out (HDFO) technology, to fabricate the interposer in Cu and organic dielectrics. This eliminates the need for an expensive interposer die. The HDFO subsystem then can be placed on the package substrate. Amkor calls this type of structure S-SWIFT® (Substrate Silicon Wafer Integration Fanout Technology) packaging.

With added functionalities, the ASIC die seem to be approaching full reticle size. Integrated with discrete HBMs and SerDes chips, some package substrates are approaching 75 x 75 mm and 85 x 85 mm sizes today, with high density modules approaching 40 x 50 mm. Looking to the future, some of the packages will include silicon photonics and will get as large as 100 mm on each side.

For data rates above 50 Gbps, pulse-amplitude modulation with four levels (PAM4) is used to lower the Nyquist frequency and reduce the channel loss. To avoid the decreased signal to noise ratio (SNR), increased power and cross-talk issues that are associated with PAM, "a highly integrated system that can bring devices closer to each other in order to reduce the interconnect distance" is the long-term solution according to SEMI's Heterogeneous Integration Roadmap, 2019 Edition[2].

Advanced Multi-layer Packaging

SWIFT packaging is a High-Density Fan-Out (HDFO) technology developed by Amkor in 2013. This design technique allows multiple dies to be assembled together with RDL techniques in increasingly smaller and tighter geometries. It is a die-last process and therefore die are committed only to known good sites after fully fabricating the fan-out layers and using exhaustive AOI techniques determine good sites available for die attach. This helps to improve yields. With smaller geometries, SWIFT design provides better performance at a lower cost. It is already widely used for mobile applications and can also be used for Networking and High-Performance Computing (HPC) applications. Primary drivers for this advanced packaging design approach include:

- Reduced form factor
- Enhanced signal integrity
- Superior impedance matching
- Optimized power distribution

Figure 3 shows the typical construction of a substrate SWIFT or S-SWIFT (HDFO on substrate) structure. Typically, S-SWIFT design has a 4 RDL (RDL-first, chip last) construction Layer 1 and Layer 3 are used for signal routing and Layer 2 is used for the ground plane. Layer 4 can be used for mixed purposes – either as a plane or for copper pillar (CuP) interconnect.

S-SWIFT packaging can support bump pitches of 30 to 80 μm (typical), with line/spacing of 2/2 μm for RLD layers 1-4. Several customers are considering the use of SWIFT technology for integrating ASIC and chiplets (SerDes, HBM, and others). With its excellent electrical properties and flexibility, SWIFT technology is also a good candidate for die partitioned modules. Figure 3 shows how the multiple layers can be interconnected.

Looking at the roadmap of Networking devices, it seems heterogeneous packaging technology can be used in many different forms (see Figure 4). Today's Networking switch is a monolithic SoC in

A20: original issuer-authored MEPTEC article, Summer 2020, page 3. Historical S-SWIFT cross-sections and networking architecture. Bandwidth, package dimensions and process limits belong to that dated example and are not carried into the financial forecast.

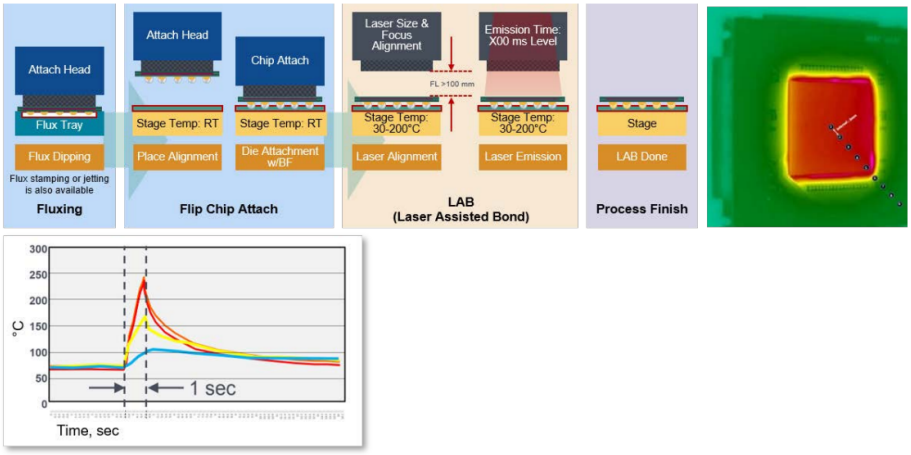


Figure 8. Laser-assisted bonding profiles in different areas in an HDFO package.

mance. Figure 5 shows a setup used by Amkor for simulation purposes. Simulations include both the signal path as well as ground and power supply noise.

As shown in Figure 6, S-SWIFT packaging shows much lower insertion loss compared to 2.5D signal routing in the interposer. Off-chip signals Through Silicon Via (TSV) packaging. With 2.5D structure, customers can expect 3 dB signal loss at around 4 GHz. The SWIFT structure can go beyond 10 GHz before hitting the 3 dB insertion loss mark. This shows the extra margin designers of an application can expect with SWIFT construction.

As shown in Figure 7, a SWIFT structure has lower losses compared to a TSV structure. These are signals going off-package.

With more functionality at higher speeds, an effective solution has a to deal with a large amount of power dissipation. It is not unheard of to have access of 500W of power that needs to be dissipated. Customers need to determine if they want their solution to be lidded or if they want to have a bare die with a stiffener ring. The TIM material that sits between the silicon and lid material or between lid and heat sink can plays a critical role in getting the heat out of the package. To determine the right choice, there are several developments underway to optimize thickness of the lid and TIM material selection, including commercialization of a lower-cost Indium metallurgical TIM.

Another challenge with large body heterogeneous packages is warpage. Warpage is unavoidable, but there are ways to limit it, to make the package yields better. One technology has helped in this area is Laser-assisted bonding (LAB). LAB Technology uses a laser to heat up the die locally and solders it to the substrate. Some of the advantages of LAB are:

- Avoids bulk heating and minimizes the coefficient of thermal expansion (CTE) mismatch between the IC and the substrate
- Provides excellent warpage control for large, thin substrates
- Means less side wall solder wicking

Figure 8 shows different aspects of using LAB technology. With local heating the die, bump and low-k layer stress levels can be lowered as the substrate expansion and contraction is minimized.

As part of our advanced packaging research, and especially for high speed networks, Amkor is working continuously to increase the number of chiplets in its designs and also increase the size of the S-SWIFT package. With HBMs moving from HBM2 to HBM2E and soon to HBM3, Amkor's close work with its customers will help us to be an integral part of this transition. Other efforts include plans to use embedded bypassing caps in new products at the module level. We are also looking into S-Connect technology to provide L/S of 1/1 for very high-density interconnects.

At Amkor, heterogeneous integration forms the basis of our most advanced packaging designs. To help customers meet the increasing performance demands of today's network systems; a variety of technologies are already offered. Ongoing development will provide even greater performance for signal transmission, power dissipation and long-term reliability while addressing the need for cost-effective packaging solutions for future networks. ◆

REFERENCES

[1] Cisco Visual Networking Index (VNI) Global and Americas/EMEAR Mobile Data Traffic Forecast, 2017-2022

[2] Heterogeneous Integration Roadmap, 2019 Edition, Chapter 2: High Performance Computing and Data Centers, IEEE.

SWIFT is a registered trademark of Amkor Technology, Inc.

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A20: original issuer-authored MEPTEC article, Summer 2020, page 5. Thermal interface and laser-assisted bonding explain possible reliability and yield benefits. The technical discussion supplies mechanism, without establishing a current customer award or margin premium.

Management's April call described constructive pricing that largely covers material inflation. The model examines the potential for scarcity pricing, but it does not equate a larger invoice with greater net profit. The separate probability ledger explains why no additional net premium is added to the already-rising computing margins. [A04]

Supercycle and TAM: separate opportunity from company capture

Horizon / mechanism	Supported evidence	Model treatment
Current computing cycle	AI training/inference, processors, networking and advanced packaging increase complexity. May management estimates roughly 24% annual computing growth across 2025-2030. [A18B, slide 12]	Use company end-market receipts and product cost assumptions. September direction is applied to the estimated company computing base, with existing growth deducted.
2026-2030 adoption	Enterprise agents, AI networking, automotive compute and regional supply requirements can widen eligible packaging and test work.	The framework's 50% adjacent-application planning weight is a research lens, not another multiplier on receipts already captured by computing and the Arizona case.
Longer-term applications	New architectures, edge AI and further regionalization can create additional qualified work. Timing, package units and customer awards remain uncertain.	The framework's 20% distant-option planning weight does not create an extra numerical sales pool. Revisit when there is a distinct eligible company wallet.

The May financial framework targets 2028 revenue of $8,500-$9,500 million and roughly $11,000 million or more in 2030. It includes Arizona Phase 1 and excludes the subsequently expanded Phase 2. The retained operating reference already sits at the top of the 2028 range and above the broad 2030 company target. New capacity cannot be added in full on top of that growth. [A18B, slides 55, 59-62]

These company targets are management forecasts. The later product mix, quarterly margins, cost structure and recognition path are model assumptions. Industry spending, customer cash advances and nominal cleanroom area remain distinct from accepted revenue.

The latest actual: a consolidated GAAP cost anchor

AMKOR TECHNOLOGY, INC.
CONSOLIDATED STATEMENTS OF INCOME
(Unaudited)

	For the Three Months Ended June 30,		For the Six Months Ended June 30,	
	2026	2025	2026	2025
	(In thousands, except per share data)			
Net sales	$ 1,897,965	$ 1,511,392	$ 3,582,666	$ 2,832,967
Cost of sales	1,579,375	1,329,495	3,025,044	2,493,487
Gross profit	318,590	181,897	557,622	339,480
Selling, general and administrative	78,904	47,922	175,891	128,330
Research and development	39,832	42,008	81,590	87,660
Total operating expenses	118,736	89,930	257,481	215,990
Operating income	199,854	91,967	300,141	123,490

A02P: preserved Q2 2026 filing, physical page 4, original statement excerpt through operating profit. Values are in USD thousands; the report converts money to millions. Filing text was checked against issuer/SEC material; the saved PDF's remote byte identity was not reverified.

Q2 2026 actual	USD million	Accounting meaning
Revenue	1,898	Reported GAAP
Manufacturing cost	1,579	Materials, labor, factory depreciation and other production costs
Gross profit	319	Revenue less manufacturing cost
Selling/general/admin	79	Includes a property-disposal gain
Research and development	40	Reported expense
Total operating expense	119	SG&A plus R&D
Operating profit	200	Gross profit less operating expense
Net income attributable to Amkor	174	After non-operating items, tax and non-controlling interests; not the valuation numerator

An approximately $21 million property gain reduces reported SG&A. Removing its recurring use gives ordinary opex of about $140 million, consistent with next-quarter direction. The forecast does not repeat the disposal gain. The prior-year comparison also included a Nanium payment, so year-over-year operating changes are not a pure measure of underlying cost productivity. [A01, A02P]

Manufacturing costs, SBC and depreciation

2. Business Segments, Product Groups and End Markets

The following table presents revenue, profit or loss and significant segment expenses for our single operating segment:

	For the Three Months Ended June 30,		For the Six Months Ended June 30,	
	2026	2025	2026	2025
	(In thousands)			
Net sales	$ 1,897,965	$ 1,511,392	$ 3,582,666	$ 2,832,967
Less:				
Materials cost of sales	997,510	799,020	1,899,283	1,491,768
Labor cost of sales	189,131	176,886	369,680	336,050
Depreciation cost of sales	163,693	144,434	319,825	284,303
Other cost of sales	229,041	209,155	436,256	381,366
Selling, general and administrative	78,904	47,922	175,891	128,330
Research and development	39,832	42,008	81,590	87,660
Interest expense	16,971	16,810	34,681	33,619
Income tax expense	28,306	28,162	40,648	32,098
Other segment items [1]	(20,292)	(8,057)	(34,023)	(19,132)
Segment net income	$ 174,869	$ 55,052	$ 258,835	$ 76,905

(1) Other segment items include interest income, foreign currency (gain) loss, net and other (income) expense.

The following tables represent other balances included in net income or that are regularly provided to the Chief Operating Decision Maker (CODM):

	For the Three Months Ended June 30,		For the Six Months Ended June 30,	
	2026	2025	2026	2025
	(In thousands)		(In thousands)	
Interest income	$ 20,172	$ 14,992	$ 36,392	$ 29,445
Depreciation and amortization expense	179,650	158,842	350,553	312,663
Capital expenditures	463,820	146,189	688,425	226,086

	June 30, 2026	December 31, 2025
	(In thousands)	
Total assets	$ 9,716,114	$ 8,136,309

The following table presents our net sales by product group:

	For the Three Months Ended June 30,		For the Six Months Ended June 30,	
	2026	2025	2026	2025
	(In thousands)		(In thousands)	
Advanced products (1)	$ 1,556,676	$ 1,228,678	$ 2,928,677	$ 2,292,295
Mainstream products (2)	341,289	282,714	653,989	540,672
Total net sales	$ 1,897,965	$ 1,511,392	$ 3,582,666	$ 2,832,967

(1) Advanced products include flip chip, memory and wafer-level processing and related test services.

(2) Mainstream products include all other wirebond packaging and related test services.

A02P: Q2 filing, physical page 12, original cost and product-group tables. The single accounting segment is explicit. Advanced products include several packaging categories and related test; separate end-market gross margins are not reported.

Q2 manufacturing bridge	USD million
Materials	998
Labor	189
Manufacturing depreciation	164

Q2 manufacturing bridge	USD million
Other manufacturing cost	229
Total manufacturing cost	1,579
Gross profit after those costs	319

Q2 stock compensation of about $8 million remains inside GAAP costs. Total depreciation and amortization of about $180 million already includes factory depreciation. Neither item is added back to operating earnings, and manufacturing depreciation is not deducted a second time. EBITDA is a separate measure and does not replace the sanctioned operating-profit numerator.

Computing, Communications, Automotive/Industrial and Consumer are end-market allocations. Their forecast gross margins are LOW-confidence analytical splits calibrated to consolidated economics. They must not be labeled disclosed accounting segments.

The latest two calls: operating evidence and exact gaps

Management event	What was read	Implication
27 April: Q1 2026	Complete official issuer-hosted 14-page LSEG transcript: prepared remarks, every Q&A exchange and closing. No audio listening claimed. [A04]	HDFO CPU ramp in Q2, meaningful Q3 contribution and continuation into 2027; new Korean test capacity and SiP migration can free room. Average utilization improved to the low 70s.
27 July: Q2 2026	Full saved MarketBeat management reproduction through closing at 39:33, including prepared remarks and five questioners. Official PDF and independent audio verification unavailable after bounded retrieval. [A23]	Eleven interposer-based packaging customers; five HDFO customers with ten engagements; four launches in each platform during 2026. Bridge-style technology is expected in 2028, with no pull-forward assumed.
9 September: management conference	Visible management Q&A reproduction through closing; official IR confirms event date. Provider transcription errors and a generic first-half footer limit certified completeness. [A22B]	Direction for computing growth into 2027; proportional Phase 2 economics; production from 2030; funding and spending timing. These statements carry explicit source qualifications.

April management described about $50-$100 million of material-related deferrals in Q1 and a similar amount in Q2. Those are shipment timing exposures, not an independent additive backlog. Memory, substrates and customer-supplied silicon can prevent lines from loading even when the assembly site has nominal capacity. [A04]

July management attributed about two-thirds of Q2 margin expansion to utilization and the remainder to mix; the Q3 improvement was expected to be predominantly mix. Communications can soften through Q4 and the first half of 2027 as SiP moves to Vietnam. The computing growth case is assessed alongside that mix and cost pressure. [A23]

The entire May Investor Day SEC text representation was preserved and read. The original large slide PDF and original slide images were unavailable. The report therefore uses the supported text evidence without recreating a source slide or claiming complete Investor Day Q&A/audio review. [A18B]

Near-term guidance and the computing receipt bridge

Q3 2026 direction	Company guidance / calibration
Revenue	$1,950-$2,050 million; model uses the $2,000 million midpoint.
Gross margin	Around 19%; exact guided endpoints and midpoint are retained in the model.
Operating expense	About $140 million.
End-market direction	Computing up about 30% sequentially; Communications down high single digits; Automotive/Industrial up mid single digits; Consumer up mid-teens.
2026 capex	$2,500-$3,000 million.
Source status	July guidance and management direction; no new October actual result is asserted. [A01, A23]

September management indicated that roughly 30% year-on-year growth in the whole Computing end market could continue into 2027. That includes multiple packaging platforms. It is one company receipt pool, rather than a separate growth increment for each technology. [A22B, management exchanges 21, 23 and 39]

FY2026 computing estimate	USD million	Basis
Q1	354	Reported company revenue x rounded 21% computing allocation
Q2	418	Reported company revenue x rounded 22% allocation
Q3 directional proxy	543	Q2 allocation x guided approximate sequential increase
Q4 reference	595	Retained model estimate; no official Q4 guide at cutoff
Estimated full year	1,909	Rounded mix and estimated later quarters; not reported annual actual
Conditional FY2027 at 30% growth	2,481	Bounded management-direction endpoint
Already represented FY2027 computing	2,380	Existing receipt pool deducted before weighting
Unweighted difference	101	Incremental eligible pool only
Weighted new FY2027 receipts	61	60% subjective probability

The consolidated Q3 product allocation is normalized to group guidance and gives computing about $537 million; the raw directional estimate above is about $543 million. That difference, rounded mix and missing Q4

guidance make the annual computing base uncertain. The 60% weight is a judgment about the incremental company outcome, not certainty about management's words.

New weighted 2027 computing receipts earn the retained 22% manufacturing gross margin, less 3% for incremental engineering/support opex. That gives about $12 million of added 2027 operating profit. Only the Q4 portion, about $3 million of profit, lies in the active Q4 2027-Q3 2028 window. The later computing reference already absorbs this cohort; it receives no second addition from 2028.

Arizona Phase 2: customer commitments and production timing

The 8 September issuer release says commitments exceeded the first phase's 33,000 square meters. Phase 2 adds 60,000 square meters, taking planned campus space to 93,000 square meters. Total campus investment rises to $12,000 million. The total includes refinements to Phase 1 as well as the larger expansion; the change in the headline budget is not purely a new Phase 2 expense. [A10]

Milestone / economics	Evidence or model status
Construction	Issuer expects Phase 2 construction to begin late in 2027 and complete by the end of 2029. [A10]
Production	September CFO commentary points to production from 2030; actual acceptance and qualification dates remain open. [A22B]
Mature Phase 1 reference	About $1,000 million annual receipts and gross margin above 30% at full scale. May targets include Phase 1. [A18B]
Proportional Phase 2 proxy	September CFO commentary endorses pro-rata economics while products and volumes are being developed. $1,000 million x 60,000÷33,000 gives about $1,818 million annual receipts at full scale. This is a management-supported analytical proxy, not Phase 2 revenue guidance. [A22B]
Good-unit qualification	No disclosed package units, ASP, yield, detailed tool set or quarterly accepted-volume schedule. Area alone does not establish effective saleable capacity.

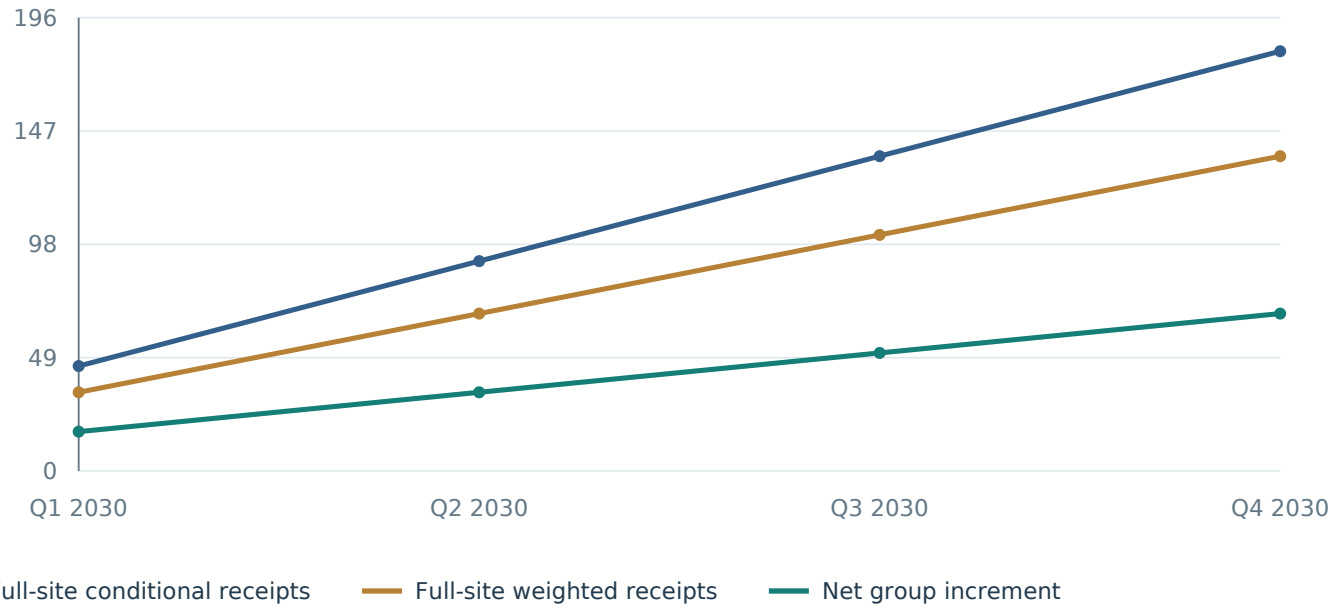
May's $7,000 million budget covers both then-planned phases. The $1,000 million revenue and greater-than-30% gross margin refer specifically to Phase 1 at full scale. September's larger $12,000 million campus includes refined Phase 1 costs and expanded Phase 2. Neither the earlier budget nor its increase is a standalone Phase 2 equipment pool. [A18B, slide 59; A22B, exchange 45]

The full-scale receipt proxy is a bounded endpoint. The model separately specifies accepted production, commissioned readiness, paid factory costs and the receipt/profit removed from the existing Computing forecast. The entire new site carries its operating burden even when only half its receipts are new to the group.

Principal driver: the full factory and the net new receipt pool

Phase 2 receipts: full site, replaced Computing and net group growth

USD million of quarterly receipts



LOW model estimates. Accepted full-site fractions are 10%, 20%, 30%, 40% of full capacity. A 75% customer-qualification weight applies to full-site receipts. Half the weighted receipts replace Computing already in the reference; net group growth is the remainder. Whole-site costs are retained, rather than halved with net receipts.

2030 quarter	Full-site conditional	Full-site weighted	Removed Computing	Net group increment	Paid factory fixed cost
Q1 2030	45	34	17	17	30
Q2 2030	91	68	34	34	43
Q3 2030	136	102	51	51	55
Q4 2030	182	136	68	68	68

The retained reference contains $12,000 million of 2030 group receipts, above the May company target of roughly $11,000 million or more. The analytical overlap assumption removes half of full-site weighted receipts from Computing and removes the corresponding 30% Computing gross profit. It releases only one-half percent of those displaced receipts as variable opex. Existing fixed costs and Phase 1 program costs remain.

The full Phase 2 site is then added with all accepted receipts, variable manufacturing costs and paid factory readiness. This replaces an unlocated forecast wallet; it does not assert that an existing physical factory closes. Group net new receipts are about $170 million in 2030, while full-site weighted receipts are about $341 million.

Conditional on the common assumed physical-readiness path, customer qualification is on time with 75% probability or delayed one year with 25% probability. Both outcomes pay the same readiness costs. Construction and commissioning delay are separate diagnostics, so these two commercial outcomes are not presented as an exhaustive distribution of every project risk.

Readiness leads accepted production and carries a fixed cost

The mature factory calibration uses a 30% gross-margin floor from the greater-than-30% Phase 1 reference. Variable manufacturing is a LOW 55% receipt proxy. The remainder is a mature fixed factory pool: full annual receipts x 15%, or about $273 million a year. That pool includes depreciation, paid manufacturing staff and other fixed readiness. It is not a disclosed asset budget.

Quarter / state-independent readiness	Accepted fraction if qualified	Average tool readiness	Full factory fixed cost, USD m
Q4 2029	0%	Shell/core floor only	17
Q1 2030	10%	25%	30
Q2 2030	20%	50%	43
Q3 2030	30%	75%	55
Q4 2030	40%	100%	68

A quarter of the mature fixed pool represents a non-scaling ready-shell/core floor. The other three-quarters follow average commissioned-tool readiness of 25%, 50%, 75% and 100% during 2030. Thus accepted output can lag physical readiness. Completing the shell at the end of 2029 also produces about $17 million of preparation expense in Q4 2029, with no accepted receipts.

Formula: quarter fixed cost = annual mature fixed pool ÷4 x [25% +75% x average tool readiness]. This gives paid 2030 factory costs of about $30 million, $43 million, $55 million and $68 million, totaling about $196 million. The readiness schedule and shell/core split are explicit LOW assumptions, not construction or tool guidance.

Full-site 2030 economics, probability weighted	Q1	Q2	Q3	Q4
Full-site receipts	34	68	102	136
Variable manufacturing cost	19	38	56	75
Qualified factory cost in COGS	22	32	42	51
Full-site gross profit	-7	-1	4	10
Full-site gross margin	-21%	-2%	4%	8%
Unqualified readiness in opex	7	11	14	17

Full-site gross margin is calculated on full-site receipts. It is negative in the first two quarters and becomes modestly positive as utilization improves. It is never divided by the smaller net group increment. The delayed unqualified outcome has zero sales and zero manufacturing GP, but its paid readiness appears in opex.

State accounting keeps preparation costs paid once

Management says depreciation and workforce preparation begin before production, initially in opex, then move into cost of sales when programs qualify. The on-time outcome therefore places qualified factory readiness in COGS. The one-year unqualified delay places the same paid burden in preproduction opex. Costs are probability-weighted after this classification, with no depreciation charged twice. [A04, pages 5 and 7; A22B, exchange 41]

2030 outcome	Full-site receipts	Removed reference GP	Full-site GP	Preop readiness opex	Net group OP increment
Qualified on time	455	68	9	0	-126
Qualification delayed one year	0	0	0	196	-236
75% /25% weighted	341	51	6	49	-153

The net OP column also deducts separate program administration and 6% support on full accepted site receipts, while releasing the existing one-half percent variable opex on displaced Computing. Paid design/program work continues in both commercial outcomes. Manufacturing staff, factory depreciation and readiness are excluded from that separate program line.

All 2030 amounts are USD millions. Full-site gross profit of about $6 million becomes a negative group GP increment after about $51 million of displaced Computing gross profit is removed. Readiness opex, full-site support and program costs then give a net 2030 operating-profit increment of about negative $153 million. This illustrates why a larger campus can initially dilute profit while building later capacity.

The cost bridge explains the nearer-window constraint

The nearer selected window precedes factory readiness. Its new Phase 2 burden is separate design, qualification and program administration. The later factory cost is explicitly tied to shell readiness, tool commissioning and accepted output.

Additional Phase 2 cost	Frozen operating treatment
Design / qualification / program administration	$5 million in Q4 2027; $10 million each quarter in 2028; $15 million each quarter in 2029; $10 million each quarter in 2030.

Additional Phase 2 cost	Frozen operating treatment
Variable support	6% of FULL accepted Phase 2 site receipts; release only the existing one-half percent variable opex on removed reference receipts.
Delayed outcome	Paid factory readiness and separate program administration continue when customer qualification is delayed.
Excluded from the program line	Manufacturing readiness staff, factory depreciation, capitalized construction and existing Phase 1 startup expense.
Evidence strength	Factory cost calibration, readiness, shell/core split and new program/support costs are LOW estimates, with no numerical quarterly Phase 2 cost guide.

In the selected Q4 2027-Q3 2028 window, new Phase 2 accepted receipts are zero while additional program expense is $35 million. The net computing contribution is about $3 million. Their combined effect is about $32 million of incremental operating cost after benefit. The larger site can strengthen the long-term opportunity while its earlier preparation constrains nearer earnings.

Phase 1 costs remain separate. Existing startup opex equals one-and-a-half percent of 2027 receipts and one-half percent in 2028. The near Q3 2026 opex is fixed at guidance. Existing reference fixed costs, variable opex and product-margin underutilization remain in place. There is no duplicate building depreciation or second use of the source's Phase 1 startup estimate.

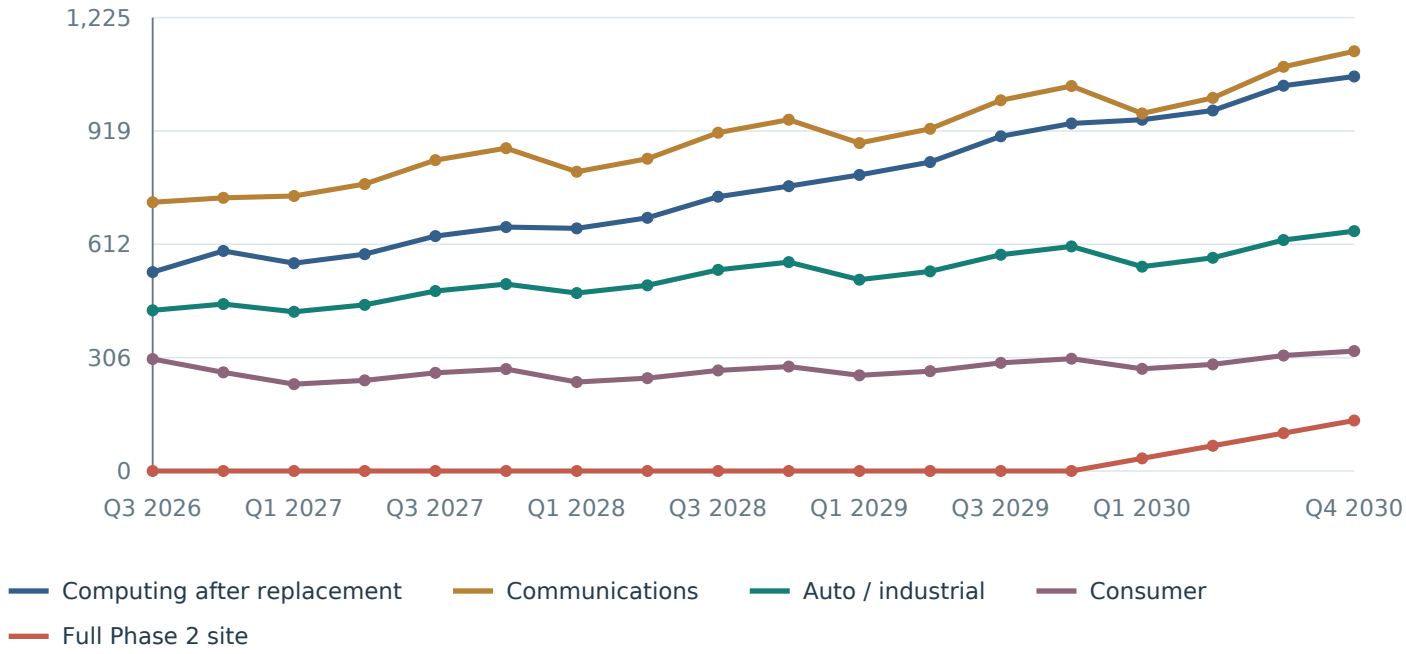
Annual new bridge	2027	2028	2029	2030
Computing receipts	61	0	0	0
Full Phase 2 site receipts	0	0	0	341
Removed Computing receipts	0	0	0	170
Net group Phase 2 receipts	0	0	0	170
Full-site gross profit	0	0	0	6
Removed Computing GP	0	0	0	51
Net group GP increment	0	0	0	-45
Phase 2 program opex	5	40	60	40
Phase 2 variable support	0	0	0	20
Preop readiness opex	0	0	17	49
Released reference variable opex	0	0	0	1
Total incremental OP	7	-40	-77	-153

Dollar amounts above are millions. Growth, probability and costs are applied to the incremental pool after existing forecasts are deducted; company capacity, customer advances and general AI spending are not added independently.

Revenue and product mix: the full quarterly operating path

Quarterly receipts by end market and the full new site

USD million; calendar quarters



Q3 2026 is guidance calibrated. Later end-market amounts are LOW model estimates. Computing is reduced by the overlapping receipt pool, then full-site probability-weighted Phase 2 receipts are added. The red site line is FULL new-site activity; its NET group increment is smaller and shown separately in the driver table.

Quarter	Net Computing	Comms	Auto / industrial	Consumer	Full Phase 2	Group total
Q3 2026	537	726	434	303	0	2,000
Q4 2026	595	738	451	267	0	2,050
Q1 2027	561	743	430	235	0	1,969
Q2 2027	586	775	449	245	0	2,055
Q3 2027	635	840	486	265	0	2,226
Q4 2027	659	872	505	275	0	2,311
Q1 2028	656	808	481	240	0	2,185
Q2 2028	684	844	502	251	0	2,280
Q3 2028	741	914	543	272	0	2,470

Quarter	Net Computing	Comms	Auto / industrial	Consumer	Full Phase 2	Group total
Q4 2028	770	949	564	282	0	2,565
Q1 2029	800	886	517	258	0	2,461
Q2 2029	835	924	539	270	0	2,568
Q3 2029	904	1,002	584	292	0	2,782
Q4 2029	939	1,040	607	303	0	2,889
Q1 2030	949	966	552	276	34	2,777
Q2 2030	974	1,008	576	288	68	2,914
Q3 2030	1,041	1,092	624	312	102	3,171
Q4 2030	1,066	1,134	648	324	136	3,308

All five vectors sum to total company receipts in every quarter. Computing reflects the matched replacement deduction; the full new-site line retains its complete receipt and cost burden. No product GP is divided by the smaller net group increment. Revenue timing requires qualified and accepted packages or services.

Every product: receipts and gross profit

This operating detail retains all eighteen forward quarters. Tables use USD millions. Product gross profit already deducts manufacturing costs, including depreciation. Whole-number display is separate from exact model arithmetic.

Quarter	Computing receipts	Computing GP	Communications receipts	Communications GP
Q3 2026	537	127	726	134
Q4 2026	595	143	738	133
Q1 2027	561	124	743	134
Q2 2027	586	129	775	140
Q3 2027	635	140	840	151
Q4 2027	659	145	872	157
Q1 2028	656	138	808	146
Q2 2028	684	144	844	152
Q3 2028	741	156	914	165
Q4 2028	770	162	949	171

Quarter	Computing receipts	Computing GP	Communications receipts	Communications GP
Q1 2029	800	216	886	177
Q2 2029	835	225	924	185
Q3 2029	904	244	1,002	200
Q4 2029	939	254	1,040	208
Q1 2030	949	285	966	198
Q2 2030	974	292	1,008	207
Q3 2030	1,041	312	1,092	224
Q4 2030	1,066	320	1,134	232

Quarter	Auto / industrial receipts	Auto / industrial GP	Consumer receipts	Consumer GP
Q3 2026	434	82	303	36
Q4 2026	451	81	267	32
Q1 2027	430	82	235	28
Q2 2027	449	85	245	29
Q3 2027	486	92	265	32
Q4 2027	505	96	275	33
Q1 2028	481	91	240	29
Q2 2028	502	95	251	30
Q3 2028	543	103	272	33
Q4 2028	564	107	282	34
Q1 2029	517	109	258	34
Q2 2029	539	113	270	35
Q3 2029	584	123	292	38
Q4 2029	607	127	303	39
Q1 2030	552	121	276	39
Q2 2030	576	127	288	40
Q3 2030	624	137	312	44
Q4 2030	648	143	324	45

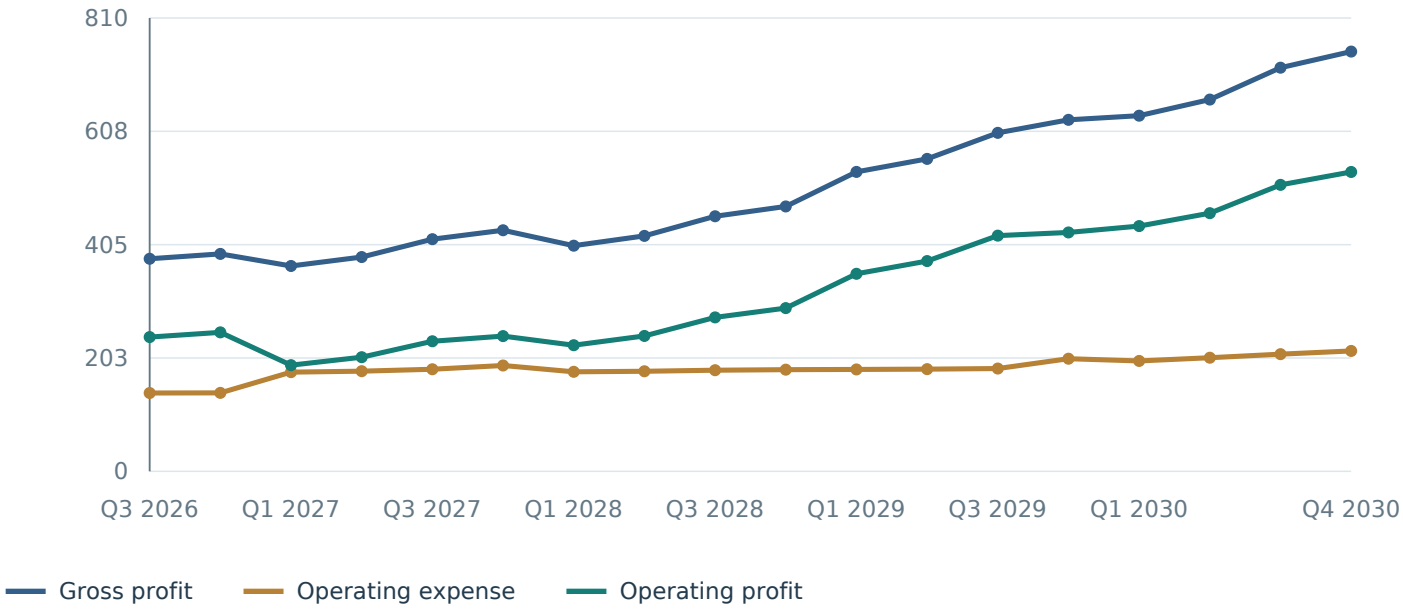
Quarter	Full Phase 2 receipts	Full Phase 2 GP	Qualified manufacturing cost
Q3 2026	0	0	0
Q4 2026	0	0	0
Q1 2027	0	0	0
Q2 2027	0	0	0
Q3 2027	0	0	0
Q4 2027	0	0	0
Q1 2028	0	0	0
Q2 2028	0	0	0
Q3 2028	0	0	0
Q4 2028	0	0	0
Q1 2029	0	0	0
Q2 2029	0	0	0
Q3 2029	0	0	0
Q4 2029	0	0	0
Q1 2030	34	-7	41
Q2 2030	68	-1	69
Q3 2030	102	4	98
Q4 2030	136	10	126

The full-site manufacturing cost equals variable manufacturing plus qualified readiness. Readiness incurred in the unqualified delay sits in opex instead. Negative early product GP reflects paid qualified capacity ahead of loaded output. The 2030 Computing tables already remove both the overlapping receipts and their 30% GP.

Profit, margins and operating costs

Quarterly gross profit, operating costs and operating profit

USD million; calendar quarters



Forward GAAP OP model. Qualified factory readiness is in COGS; unqualified readiness is in opex. Manufacturing, full reference expense and incremental program/support costs are deducted once. Both margins use the same consolidated group receipts.

Quarter	Gross profit	Operating expense	Operating profit	Gross margin	OP margin
Q3 2026	380	140	240	19%	12%
Q4 2026	389	140	248	19%	12%
Q1 2027	367	177	190	19%	10%
Q2 2027	383	179	204	19%	10%
Q3 2027	415	182	233	19%	10%
Q4 2027	431	189	242	19%	10%
Q1 2028	403	178	225	18%	10%
Q2 2028	421	179	242	18%	11%
Q3 2028	456	181	275	18%	11%
Q4 2028	473	182	292	18%	11%
Q1 2029	535	182	353	22%	14%

Quarter	Gross profit	Operating expense	Operating profit	Gross margin	OP margin
Q2 2029	559	183	376	22%	15%
Q3 2029	605	184	421	22%	15%
Q4 2029	628	201	427	22%	15%
Q1 2030	636	197	438	23%	16%
Q2 2030	665	203	461	23%	16%
Q3 2030	722	209	512	23%	16%
Q4 2030	750	215	535	23%	16%

Annual operating case	Receipts	Gross profit	Opex	OP	OP margin
2027	8,561	1,596	728	868	10%
2028	9,500	1,754	719	1,034	11%
2029	10,700	2,327	750	1,577	15%
2030	12,170	2,772	825	1,947	16%

Later margin expansion requires mix, utilization and qualified manufacturing performance. It is a model condition, not evidence that announced capex generates an automatic profit yield. Customer/material cost pass-through can increase receipts without increasing net profit at the same rate.

Quarterly cost transparency

Quarter	Reference opex	Compute support	P2 program	P2 support	Preop readiness	Released old variable	Total opex
Q3 2026	140	0	0	0	0	0	140
Q4 2026	140	0	0	0	0	0	140
Q1 2027	177	less than 1	0	0	0	0	177
Q2 2027	179	less than 1	0	0	0	0	179
Q3 2027	182	less than 1	0	0	0	0	182
Q4 2027	184	less than 1	5	0	0	0	189
Q1 2028	168	0	10	0	0	0	178
Q2 2028	169	0	10	0	0	0	179
Q3 2028	171	0	10	0	0	0	181
Q4 2028	172	0	10	0	0	0	182
Q1 2029	167	0	15	0	0	0	182
Q2 2029	168	0	15	0	0	0	183
Q3 2029	169	0	15	0	0	0	184
Q4 2029	169	0	15	0	17	0	201
Q1 2030	178	0	10	2	7	less than 1	197
Q2 2030	179	0	10	4	11	less than 1	203
Q3 2030	180	0	10	6	14	less than 1	209
Q4 2030	180	0	10	8	17	less than 1	215

Opex equals reference plus computing support, Phase 2 program/support and preproduction readiness, less the released old variable cost. The reference starts with $130 million fixed expense in 2026, grows that fixed amount 6% a year, and adds one-half percent of receipts plus existing Phase 1 startup. Q3 2026 is overridden to the guided $140 million. The release is restricted to the matched displaced receipt pool; old fixed expenses remain.

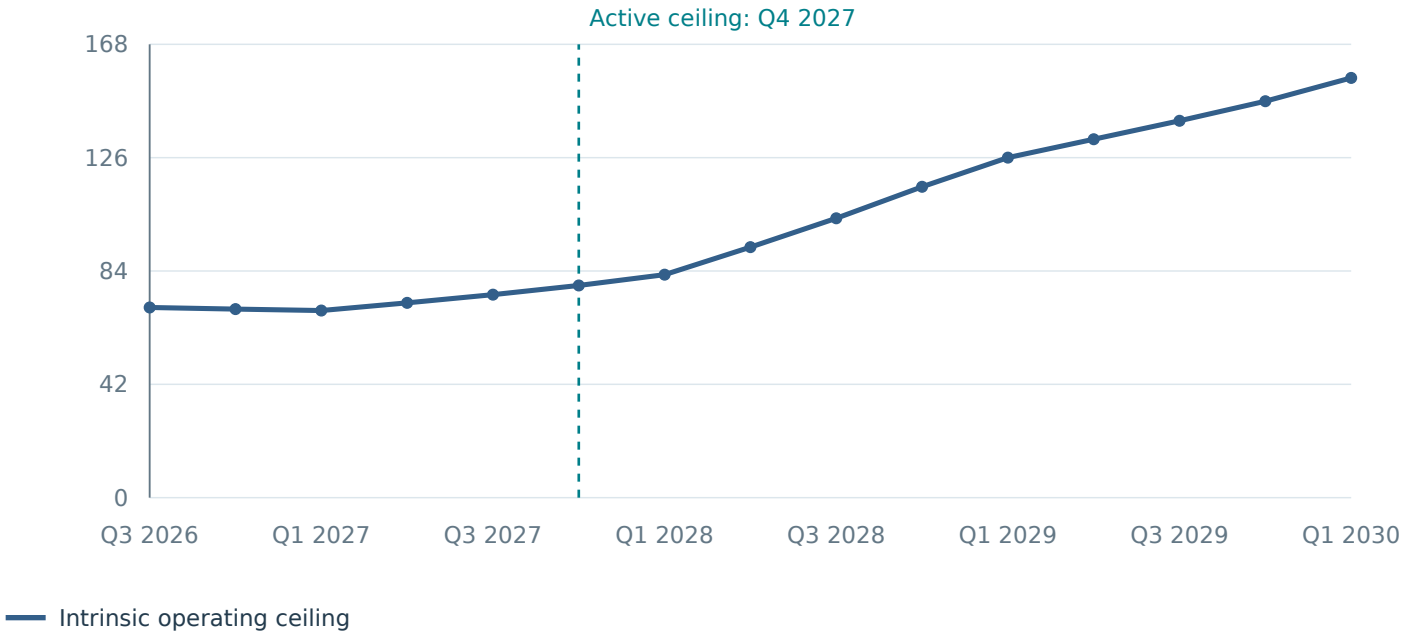
Reference end-market gross margins, rounded here to whole percentages: computing is 22% in 2027, 21% in 2028, 27% in 2029 and 30% in 2030. Communications, automotive and consumer retain their own cost paths. The full product receipt/GP tables allow those splits to be checked without implying that they are reported segment margins.

Valuation: one method across fifteen explicit windows

The sanctioned method is 20x forward-four-quarter GAAP operating profit divided by the starting-quarter diluted-share proxy. All fifteen selected quarters retain 249,978,000 shares. Financial statements are converted from thousands into millions once; OP in millions is multiplied by one million when dividing by the whole-share count. There is no ADR, split or currency conversion.

Operating ceiling path across all valued quarters

USD per common share



Model ceilings, not promised market prices. Each observation values the four quarters beginning with the labeled quarter using 20x OP and the preserved share denominator. Q4 2027 is the active workbook horizon.

Ceiling quarter	Valued four-quarter window	F4Q OP, USD m	Multiple	Shares	Ceiling, USD
Q3 2026	Q3 2026 to Q2 2027	882	20x	249,978,000	71
Q4 2026	Q4 2026 to Q3 2027	875	20x	249,978,000	70
Q1 2027	Q1 2027 to Q4 2027	868	20x	249,978,000	69
Q2 2027	Q2 2027 to Q1 2028	904	20x	249,978,000	72
Q3 2027	Q3 2027 to Q2 2028	942	20x	249,978,000	75
Q4 2027	Q4 2027 to Q3 2028	984	20x	249,978,000	79
Q1 2028	Q1 2028 to Q4 2028	1,034	20x	249,978,000	83
Q2 2028	Q2 2028 to Q1 2029	1,162	20x	249,978,000	93

Ceiling quarter	Valued four-quarter window	F4Q OP, USD m	Multiple	Shares	Ceiling, USD
Q3 2028	Q3 2028 to Q2 2029	1,296	20x	249,978,000	104
Q4 2028	Q4 2028 to Q3 2029	1,442	20x	249,978,000	115
Q1 2029	Q1 2029 to Q4 2029	1,577	20x	249,978,000	126
Q2 2029	Q2 2029 to Q1 2030	1,663	20x	249,978,000	133
Q3 2029	Q3 2029 to Q2 2030	1,748	20x	249,978,000	140
Q4 2029	Q4 2029 to Q3 2030	1,839	20x	249,978,000	147
Q1 2030	Q1 2030 to Q4 2030	1,947	20x	249,978,000	156

This page shows the current intrinsic ceiling path. It uses no stock-price overlay, analyst target, enterprise-value substitution or ranking. Later ceiling growth depends on the operating assumptions already presented, including qualification, utilization, costs and customer acceptance.

The fifteen four-term operating-profit chains

Every chain uses the same frozen quarterly OP values shown in the operating table. OP terms and sums below are USD millions rounded for readability. The exact model calculates each sum before display rounding. Formula for every row: sum of the four quarterly OP terms x 20 x 1,000,000 ÷249,978,000 = ceiling in USD per share.

Ceiling quarter	Four quarterly OP terms, USD million	F4Q OP	Multiple	Whole shares	Ceiling
Q3 2026	240 + 248 + 190 + 204 Q3 2026 to Q2 2027	882	20x	249,978,000	71
Q4 2026	248 + 190 + 204 + 233 Q4 2026 to Q3 2027	875	20x	249,978,000	70
Q1 2027	190 + 204 + 233 + 242 Q1 2027 to Q4 2027	868	20x	249,978,000	69
Q2 2027	204 + 233 + 242 + 225 Q2 2027 to Q1 2028	904	20x	249,978,000	72
Q3 2027	233 + 242 + 225 + 242 Q3 2027 to Q2 2028	942	20x	249,978,000	75
Q4 2027	242 + 225 + 242 + 275 Q4 2027 to Q3 2028	984	20x	249,978,000	79
Q1 2028	225 + 242 + 275 + 292 Q1 2028 to Q4 2028	1,034	20x	249,978,000	83

Ceiling quarter	Four quarterly OP terms, USD million	F4Q OP	Multiple	Whole shares	Ceiling
Q2 2028	242 + 275 + 292 + 353 Q2 2028 to Q1 2029	1,162	20x	249,978,000	93
Q3 2028	275 + 292 + 353 + 376 Q3 2028 to Q2 2029	1,296	20x	249,978,000	104
Q4 2028	292 + 353 + 376 + 421 Q4 2028 to Q3 2029	1,442	20x	249,978,000	115
Q1 2029	353 + 376 + 421 + 427 Q1 2029 to Q4 2029	1,577	20x	249,978,000	126
Q2 2029	376 + 421 + 427 + 438 Q2 2029 to Q1 2030	1,663	20x	249,978,000	133
Q3 2029	421 + 427 + 438 + 461 Q3 2029 to Q2 2030	1,748	20x	249,978,000	140
Q4 2029	427 + 438 + 461 + 512 Q4 2029 to Q3 2030	1,839	20x	249,978,000	147
Q1 2030	438 + 461 + 512 + 535 Q1 2030 to Q4 2030	1,947	20x	249,978,000	156

The active calculation values Q4 2027 plus Q1, Q2 and Q3 2028. Its four terms total about $984 million; the resulting operating ceiling rounds to $79 per share. The 20x method, OP accounting basis and share convention are preserved rather than recalibrated to a desired stock price.

Probability ledger: active increments and justified zero additions

Evidence / probability	Conditional bridge and overlap	Active treatment
September computing direction: 60%	Estimated FY2026 computing base x 30% growth; deduct the existing $2,380 million 2027 receipt pool; allocate the remaining gap across four quarters. Apply 22% manufacturing GP and deduct 3% support opex. Later cohorts are fully absorbed in the reference.	About $61 million new 2027 receipts and $12 million OP; about $3 million OP in the active four-quarter window.
New Phase 2: 75% qualification on time /25% one-year delay	Full-site receipt proxy about $1,818 million annual at maturity. Accepted ramp 10/20/30/40%, separate 25/50/75/100% readiness and 25% fixed floor. Remove matched Computing receipts/GP before adding the full new site. Both outcomes pay readiness.	No accepted sales before 2030. Active window has $35 million program costs. In 2030 the full site earns about $341 million weighted receipts, of which $170 million is net new to the group; net OP increment is about negative $153 million after full costs.
Q2 actual: 100%	Exact group financial, manufacturing-cost and share anchor; remove recurring use of the property gain.	Calibration already included; no second addition to sales or OP.
Q3 guidance: 90%	Revenue/margin midpoint and recurring opex anchor already populate the near-quarter bridge.	No second guidance pool.
Korea HDFO/CPU and later bridge products: 65%	Existing computing receipts and margin path already require platform ramp and qualification; bridge timing remains 2028.	No separate technology multiplier.
Phase 1 ramp: 75%	Existing startup opex and margin/underutilization assumptions retain its 2027-2030 burden.	No repeated use of Phase 1 costs or Phase 1 full-scale revenue.
NVIDIA advance: 85%	Expected customer funding is credited against services over five to ten years.	No immediate profit or additional annual sales pool.
TSMC collaboration: 80%	Ten-year procurement relationship supports qualification and regional capacity; volumes, ASP and exclusivity are not disclosed.	Included in the capacity/reference opportunity; no unbounded global award.
Positive new net pricing: 30%	Attempt a premium on eligible advanced-compute contracts, subtract material inflation and support. Existing computing gross-margin improvements already require better mix, utilization and cost recovery.	100% overlap with the current net-margin pool; zero extra OP. This is a positive outcome probability with no additional unmodeled pool, rather than a confirmation-only exclusion.
SpaceX spending read-through: 35%	Attempt the chain from compute deployment to eligible OSAT wallet, Amkor capture, accepted packages, price and timing. No distinct new Amkor award/unit/ASP pool was identified.	Current computing and both Arizona layers already cover the plausible qualified wallet. Zero separate macro-capex addition; revisit a distinct company receipt pool.

All probabilities are model judgments about conditional company outcomes. They are not additive market shares. Separate stages of a customer program, overlapping end markets, advances and construction commitments are reconciled before any addition. Zero increments above state their economic or overlap reason.

Sensitivity: timing, overlap and full factory economics

All twenty-five diagnostics recalculate the same operating model, retaining 20x OP and the share convention. The named input changes while unmentioned inputs stay fixed. These are analytical diagnostics and controls, not separate approved price targets. Program/support cost sensitivities leave factory readiness and computing support unchanged; factory-fixed tests vary a different cost pool.

Changed input / diagnostic	Q4 2027 ceiling, USD	Q1 2030 ceiling, USD
No extra computing growth (0% weight)	79	156
Full computing endpoint (100% weight)	79	156
All customer qualification delayed; readiness paid	79	149
All customer qualification on time	79	158
No net new group receipts; whole replacement costs retained	79	152
No receipt overlap deduction	79	160
Double initial accepted receipts	79	162
Zero Phase 2 program/support only	82	161
Double Phase 2 program/support only	76	151
Zero paid factory fixed cost	79	171
Double paid factory fixed cost	79	140
No shell/core fixed floor	79	158
Half shell/core fixed floor	79	154
Full capacity ready throughout 2030	79	150
Construction, readiness and acceptance one year later	79	163
Variable COGS 45%; fixed recalibrated to 30% mature GM	79	148
Variable COGS 60%; fixed recalibrated to 30% mature GM	79	160
Mature full-scale gross margin 35%	79	161
Mature full-scale gross margin 40%	79	166

Changed input / diagnostic	Q4 2027 ceiling, USD	Q1 2030 ceiling, USD
Low fixed burden: variable 60%, shell floor 10%	79	160
Heavy fixed burden: variable 45%, shell floor 50%	79	145
Extra $1,000 million net equipment, seven-year life	79	148
Extra $3,000 million net equipment, seven-year life	79	131
No new hypotheses: whole Phase 2 investment disabled	81	168
Note/capped-call net above-cap share-equivalent diagnostic	Not applied	155

The one-year construction/readiness delay produces a higher 2030-window ceiling because it defers the initial loss-making readiness and ramp costs. Accepted receipts and capacity also shift beyond 2030. This is a short-window expense deferral effect; it does not establish that delaying the project improves its economic value.

The no-net-new-receipts test removes all incremental group demand but keeps the full new-site operation, readiness and replacement cost. The no-new-hypotheses control instead disables the entire Phase 2 investment and the separate computing hypothesis for a calculation-recovery check. These controls answer different questions and are not interchangeable.

The extra equipment tests add annual depreciation on assumed net commissioned equipment after other basis adjustments: $1,000 million ÷7 or $3,000 million ÷7, phased through the readiness path. They expose unallocated-capital risk. They do not assert an actual equipment purchase, claim all campus capital has been allocated, or replace the operating case.

Mature-margin and variable-cost tests recalibrate the fixed pool to preserve the specified mature economics; they are not independent free-margin additions. Independent validation checks all quarter/product/cost/state joins and every four-quarter chain. Forecast and source uncertainty remain.

Cash, working capital and capital commitments

(Unaudited)

	June 30, 2026	December 31, 2025
	(In thousands, except per share data)	
ASSETS		
Current assets:		
Cash and cash equivalents	$ 1,551,890	$ 1,378,347
Restricted cash	1,580	—
Short-term investments (amortized cost of $962,740 and $612,344 in 2026 and 2025, respectively)	960,272	613,038
Accounts receivable, net of allowances	1,420,353	1,354,825
Inventories	560,681	437,797
Other current assets	113,853	100,754
Total current assets	4,608,629	3,884,761
Property, plant and equipment, net	4,478,951	3,870,808
Operating lease right of use assets	84,440	93,449
Goodwill	17,356	18,003
Restricted cash	186,875	67,776
Other assets	339,863	201,512
Total assets	$ 9,716,114	$ 8,136,309
LIABILITIES AND EQUITY		
Current liabilities:		
Short-term borrowings and current portion of long-term debt	$ 150,781	$ 162,430
Trade accounts payable	880,686	912,766
Capital expenditures payable	621,326	243,543
Short-term operating lease liability	22,766	23,140
Accrued expenses	447,291	370,093
Total current liabilities	2,122,850	1,711,972
Long-term debt	2,334,950	1,282,816
Pension and severance obligations	68,127	69,218
Long-term operating lease liabilities	38,943	48,549
Other non-current liabilities	464,416	517,467
Total liabilities	5,029,286	3,630,022
Commitments and contingencies (Note 15)		
Stockholders' equity:		
Preferred stock, $0.001 par value, 10,000 shares authorized, designated Series A, none issued	—	—
Common stock, $0.001 par value, 500,000 shares authorized; 294,562 and 293,756 shares issued; and 247,986 and 247,303 shares outstanding in 2026 and 2025, respectively	295	294
Additional paid-in capital	2,025,469	2,054,051
Retained earnings	2,842,719	2,627,038
Accumulated other comprehensive income (loss)	15,107	16,833
Treasury stock, at cost, 46,576 and 46,453 shares in 2026 and 2025, respectively	(233,178)	(227,110)
Total Amkor stockholders' equity	4,650,412	4,471,106
Non-controlling interests in subsidiaries	36,416	35,181
Total equity	4,686,828	4,506,287
Total liabilities and equity	$ 9,716,114	$ 8,136,309

The accompanying notes are an integral part of these statements.

A02P: preserved Q2 filing, physical page 6. Original consolidated balance sheet excerpt. Current and noncurrent restricted cash are separate from ordinary cash; capex payables are explicit obligations.

Latest financial position	USD million	Interpretation
Cash and short-term investments	2,512	Ordinary cash plus investments; restricted cash excluded
Restricted cash	188	Current and noncurrent combined; not general excess cash
Debt carrying amount	2,486	Current plus long-term balance after debt costs
Debt principal	2,516	Different basis from carrying debt
Capital-expenditure payables	621	Equipment/building cash obligations remain
Receivables	1,420	Customer cash conversion exposure
Inventory	561	Loading and materials exposure
Purchase/other commitments due within a year	1,395	Limits available financial flexibility
First-half operating cash flow	382	Reported cash from operations
First-half cash capex	688	Investment uses cash before mature earnings
Issuer-defined first-half free cash flow	-270	Includes about $37 million of PPE sale/grant proceeds in its reconciliation

The operating ceiling does not net cash or debt into a different valuation method. Funding nevertheless matters to execution and share value. The company has an undrawn $1,000 million revolver and about $55 million of other credit availability, subject to terms. Financing availability is not revenue or operating profit. [A02P]

Customer cash, incentives and the campus funding gap

We had off-balance sheet purchase obligations for capital expenditures, long-term supply contracts and other contractual commitments. As of June 30, 2026, the purchase obligations were $1,465.2 million, with $1,395.1 million payable within 12 months.

We enter into customer advance payment agreements from time to time, some of which require standby letters of credit. As of June 30, 2026, we expect to receive approximately $300 million of advance payments over a two-year period, all of which will require standby letters of credit upon receipt. In July 2026, we received $100.0 million of customer advance payments and issued the related standby letters of credit. During the same month, we separately entered into an advance payment agreement with a customer pursuant to which we expect to receive approximately $1.5 billion in 2027. For additional information regarding our customer advance payments, please refer to Note 1 to our Consolidated Financial Statements in Part I, Item 1 of this Form 10-Q.

Capital Returns

In November 2022, we announced our intention to return 40 percent to 50 percent of cumulative free cash flow generated over time, beginning 2022. This return may be in the form of dividends and stock repurchases, subject to a variety of factors, including strategic investments, other capital allocation priorities and Board of Directors' approval.

During the six months ended June 30, 2026, we paid total quarterly cash dividends of $41.4 million, and we currently anticipate that we will continue to pay quarterly cash dividends in the future. However, the payment, amount and timing of future dividends remain within the discretion of our Board of Directors and will depend upon our results of operations, financial condition, cash requirements, debt restrictions and other factors.

On April 23, 2026, our Board of Directors adopted a stock repurchase program (the "Stock Repurchase Program") authorizing the repurchase of up to $300.0 million of our common stock, exclusive of any fees, commissions or other expenses. Under the Stock Repurchase Program, the purchase of stock may be made in the open market or through privately negotiated transactions. The timing, manner, price and amount of any repurchases will be determined by us at our discretion and will depend upon a variety of factors including economic and market conditions, the cash needs and investment opportunities for the business, the current market price of our stock, applicable legal requirements and other factors. At June 30, 2026, $300.0 million was available to repurchase common stock pursuant to the Stock Repurchase Program.

A02P: Q2 filing, physical page 32, original upper excerpt. Customer advances finance later obligations; capital returns are discretionary. Exact source values and language are retained.

NVIDIA's approximately $1,500 million advance is expected in 2027 and is credited against US services over an expected five to ten years. It helps finance capacity while creating a delivery obligation. It is not immediate operating profit. Q2 contract liabilities were $393 million before that future payment. [A11; A02P]

Capital Resources

We make significant capital expenditures in order to service the demand of our customers. During the six months ended June 30, 2026, our capital expenditures totaled $688.4 million, which are primarily focused on investments in advanced packaging and test equipment and the Arizona Facility.

We expect that our 2026 capital expenditures will be approximately $2.5 billion to $3.0 billion. The increase from 2025 is primarily due to the construction of the Arizona Facility. Ultimately, the amount of our 2026 capital expenditures will depend on several factors including, among others, the timing and implementation of any capital projects under review, including the progress of construction of the Arizona Facility, the performance of our business, economic and market conditions, the cash needs and investment opportunities for the business, the need for additional capacity to service anticipated customer demand, equipment lead times and the availability of cash flows from operations or financing. The primary sources of funds for our capital expenditures are cash flows from operations, current cash and cash equivalents, short-term investments, borrowings under available credit facilities and proceeds from any additional debt or equity financings. Please refer to Note 7 and Note 11 to our Consolidated Financial Statements in Part 1, Item 1 of this Form 10-Q for additional information on our investments and borrowings, respectively.

In addition, we are subject to risks associated with our capital expenditures, including those discussed in the "Risk Factors" section in Part II, Item 1A of this Form 10-Q under the caption "We make substantial investments in equipment and facilities to support the demand of our customers, which may materially and adversely affect our business if the demand of our customers does not develop as we expect or is adversely affected."

A02P: Q2 filing, physical page 32, original capital-resources excerpt. Annual 2026 investment guidance and the sources and risks of funding are shown directly. This July disclosure predates the September campus expansion.

Up to $407 million of CHIPS support and the 35% credit on qualifying US investment can help. No CHIPS payment had been received at Q2. A separate roughly $100 million Korean grant was conditional and unreceived. The entire $12,000 million campus budget is not automatically eligible for every incentive. [A02P]

September management described overlapping Phase 1 equipment spending and Phase 2 construction, with Arizona outlays peaking in 2028-2029. Its paired-year spending discussion is directional rather than an exact annual capex schedule. Funding can combine operations, customer participation, grants, credits, borrowing and capital markets; future terms remain uncertain. [A22B]

What the fixed-cost calibration covers and leaves unresolved

The primary annual-report policy uses straight-line lives of 10-40 years for buildings/improvements and 2-7 years for machinery/equipment. The test-equipment life was extended from five to seven years in 2024. April management separately discussed a seven-year equipment cycle. These lives distinguish the building shell from tools; the entire campus budget is not depreciated over one common life. [A09, Note 1; A04, page 7]

The normal factory fixed pool is about $273 million annually at full scale. If every dollar of that pool were seven-year equipment depreciation, it could support at most about $1,909 million of depreciable equipment. That is a mathematical upper bound, not an equipment budget or estimate. Buildings, paid staff, utilities, maintenance and shorter-life machinery consume part of the same cost pool and reduce the permitted equipment basis.

Capital / cost scope	Treatment and limit
Full fixed manufacturing pool	$273 million at mature receipt scale, calibrated from 30% gross margin and 55% variable manufacturing. The source supplies the mature margin reference, not this cost split.
Normal readiness	25% shell/core floor plus a staged tool-readiness path; $17 million preproduction cost in Q4 2029 and about $196 million paid factory cost in 2030.
Unallocated campus capital	No standalone Phase 2 net depreciable asset budget is disclosed. Phase 1, shared shell/infrastructure, phased tools and future assets remain unallocated. This is an explicit funding and future commissioning gap.
Credits and grants	Recognized qualifying support reduces the carrying basis of the related PPE. The entire $12,000 million campus is not automatically eligible, recognized or paid. [A02P, Note 1]
Extra equipment stress	Additional net equipment basis of $1,000 million or $3,000 million at a seven-year life tests the risk that the calibrated factory pool understates committed assets. These are stress assumptions, not disclosed purchases.

Current consolidated materials are about 53% of revenue, labor 10%, factory depreciation 9% and other manufacturing 12%. They provide a cost calibration, while new US automation, package mix, good-unit yield and wage/utility costs can differ. The 55% variable-cost choice and the physical-readiness split therefore remain LOW estimates. Unassigned capital is neither declared cost-free nor expensed in full immediately. [A02P, manufacturing-cost note]

Shares, stock compensation and the 2031 convertible

5. Earnings Per Share

Basic earnings per share ("EPS") is computed by dividing net income attributable to Amkor common stockholders by the weighted-average number of common shares outstanding during the period. The weighted-average number of common shares outstanding is reduced for treasury stock.

Diluted EPS is computed based on the weighted-average number of common shares outstanding plus the effect of dilutive potential common shares outstanding during the period. Dilutive potential common shares include outstanding stock options, performance-vested restricted stock units ("PSUs"), time-vested restricted stock units ("RSUs") and shares associated with the conversion of debt instruments.

The following table summarizes the computation of basic and diluted EPS:

	For the Three Months Ended June 30,		For the Six Months Ended June 30,	
	2026	2025	2026	2025
	(In thousands, except per share data)			
Net income attributable to Amkor common stockholders	$ 173,753	$ 54,417	$ 257,104	$ 75,545
Weighted-average number of common shares outstanding - basic	247,912	247,090	247,732	246,973
Effect of dilutive securities:				
Share-based awards	2,066	716	2,098	869
Weighted-average number of common shares outstanding - diluted	249,978	247,806	249,830	247,842
Net income attributable to Amkor per common share:				
Basic	$ 0.70	$ 0.22	$ 1.04	$ 0.31
Diluted	0.70	0.22	1.03	0.30

The following table summarizes the potential shares of common stock that were excluded from diluted EPS because the effect of including these potential shares was anti-dilutive:

	For the Three Months Ended June 30,		For the Six Months Ended June 30,	
	2026	2025	2026	2025
	(In thousands)			
Share-based awards	—	984	8	378

There were no potentially dilutive common shares issuable pursuant to the $1.15 billion aggregate principal amount of our 0.00% convertible senior notes due 2031 (the "2031 Notes") for the three and six months ended June 30, 2026, as the average market price of our common stock did not exceed the applicable conversion price during those periods. Refer to Note 11 for additional information on the 2031 Notes.

A02P: Q2 filing, physical page 14. Original share reconciliation. The denominator is diluted weighted-average shares; end-date outstanding shares and potential note shares are different measures.

The retained denominator is 249,978,000 diluted weighted-average shares: 247,912,000 basic shares plus 2,066,000 awards. Quarter-end outstanding shares were 247,986,000, and the July share snapshot was 248,452,472. No ADR or split conversion applies. The flat forward proxy is an existing convention and not a promise that future awards have no economic cost.

Q2 and first-half SBC remain in GAAP expense. The $300 million repurchase authorization was unused at Q2, and September management described modest buybacks intended to offset SBC. The forecast does not assume a large, uncommitted share retirement.

Holders may convert their notes at their option under the following conditions: (1) during any calendar quarter commencing after the calendar quarter ending on June 30, 2026 and only during such fiscal quarter, if the last reported sale price of our common stock for at least 20 trading days (whether or not consecutive) during a period of 30 consecutive trading days ending on, and including, the last trading day of the immediately preceding calendar quarter exceeds (x) before July 15, 2030, 150% and (y) on or after July 15, 2030, 130%, in each case, of the conversion price on each applicable trading day; (2) during the five consecutive business days immediately after any ten consecutive trading day period in which the trading price per $1,000 principal amount of notes for each trading day of such five consecutive trading day period was less than 98% of the product of the last reported sale price of our common stock and the conversion rate on each such trading day; (3) if we call any notes for redemption; or (4) upon the occurrence of specified corporate events. On or after April 15, 2031, until the close of business on the second scheduled trading day immediately preceding the maturity date, holders of the 2031 Notes may convert all or any portion of their notes at any time, in integral multiples of $1,000 principal amount, at the option of the holder regardless of the foregoing conditions. Upon conversion, we will settle conversions of the 2031 Notes by paying cash up to the aggregate principal amount of the 2031 Notes and pay or deliver, as applicable, cash or shares of our common stock, based on the applicable conversion rate(s).

As of June 30, 2026, none of the conditions permitting the holders of the 2031 Notes to convert their notes early had been met. Therefore, the 2031 Notes are classified as long-term debt.

A02P: Q2 filing, physical page 20, original conversion paragraphs. Full early-conversion conditions, mandatory cash settlement of principal and latest eligibility are shown. Above-principal value can use cash or shares; early conversion conditions were not met at Q2.

Capped Calls can be settled in cash or shares at our option and are expected generally to reduce the potential dilution to common stock upon any conversion of the 2031 Notes and/or offset any cash payments we are required to make in excess of the principal amount of the 2031 Notes. The Capped Calls have an initial strike price of approximately $106.37 per share and an initial cap price of $139.50 per share, which are subject to certain adjustments under the terms of the Capped Calls. The Capped Calls meet the criteria for classification in equity, are not remeasured each reporting period and are included, net of tax, as a reduction to additional paid-in-capital within shareholders' equity in the Consolidated Balance Sheets.

A02P: Q2 filing, physical page 21, original opening paragraph. Exact strike, cap, cash/share settlement and accounting language for the capped call are retained. Human prose rounds financial figures, while this original source exhibit preserves its decimals.

The notes have $1,150 million principal, zero coupon and July 2031 maturity. Their original conversion terms correspond to 10,811,495 underlying shares. Full principal is not added to the forecast share count. The capped call reduces above-principal exposure up to an approximate $140 cap, subject to its terms.

A separate illustrative terminal diagnostic solves residual above-cap share-equivalent exposure while retaining cash principal settlement. The Q1 2030 ceiling rounds to $155 under that diagnostic; the current operating case for that same terminal quarter rounds to $156. It ignores tax, cash-settlement alternatives and possible term changes. This remains a diagnostic rather than an active share-denominator change or a claim that conversion conditions are already satisfied.

Competition, counter-evidence and failure conditions

Risk / counter-evidence	Why it matters	Evidence to monitor
Alternative qualified supply	ASE, JCET, Powertech, foundries, electronics manufacturers and customer in-house packaging compete. A regional advantage does not imply a permanent global monopoly. [A09]	Program allocation, customer design wins, relative yield, cycle time and price.
Commercial commitments and acceptance	The campus release improves visibility, while the filing describes short customer commitments and no broad company backlog. Capacity commitment is not the same as accepted shipment. [A10; A02P]	Firm commercial scope, acceptance dates, customer concentration and cancellations.
Factory yield and utilization	Fixed manufacturing cost and depreciation persist when a new process ramps slowly. Larger or more complex packages reduce effective good-unit capacity.	Qualified yield, mix, good-package throughput and equipment utilization.
Material and silicon bottlenecks	Missing customer-supplied silicon, memory or substrates can defer loading even where Amkor has available space. [A04]	Matched components, customer wafer readiness and receivable/inventory conversion.
Mobile / SiP migration	Communications remains material and can soften as programs move between countries. [A23]	Q4 and first-half 2027 Communications receipts, Vietnam ramp and Korean reallocation.
Funding, awards and note exposure	Heavy investment precedes mature cash generation. Borrowing, equity terms, SBC and above-cap note settlement can affect shareholder capture.	Free cash flow, customer prepay receipt, grant/credit qualification and capital-payable settlement.
Factory and program cost proxy	No official standalone Phase 2 commissioned-asset budget or quarterly operating-cost guide supports the exact readiness and program split. Unallocated capital can make the assumed factory pool too small.	Equipment cohort, staffing, qualification and credits; distinguish production costs, program administration and cash capex.

The customer-qualification delay carries the common physical-readiness cost before receipts. Construction delay instead shifts both readiness and accepted capacity, and is tested separately. Weaker net margins, lower customer allocation or more commissioned assets can reduce profit even if the building completes. Independent competing capacity and substitution matter alongside growing industry demand.

Scorecard and catalysts

Category	Points	Maximum	Reason
Core thesis	15	20	Qualified manufacturing, test and customer relationships; strong competing OSAT/foundry options.
Cycle stage	8	10	Computing programs and utilization improve, while mobile migration and startup costs persist.
TAM expansion	8	10	Expanded regional opportunity with qualification, cost and existing forecast overlap.
Catalysts	5	5	October earnings, HDFO/CPU program conversion, site and qualification milestones.
Macro	9	10	AI complexity and regional supply-chain requirements support demand.
Alpha	3	5	Recognized receipts, net costs and timing provide a more useful earnings bridge than headline campus spending.
Risks	-5	0	Capital/FCF, qualification/yield, competing allocation and mobile concentration each deduct one point. Convert/SBC and additional campus engineering each deduct one-half point, avoiding a duplicate funding/construction penalty.
Raw total	43	60	Rounded analytical score: 72%. Protected Excel Base: 72.

Catalyst	What would change confidence
26 October 2026 earnings	Actual computing mix, net price recovery, Q4 loading, material availability, cash capex and funding. This event is future at the research cutoff.
Korea / Vietnam reallocation	Confirmed productive test capacity and accepted HDFO/CPU work without duplicate counting of moved SiP receipts.
Late-2027 Phase 2 construction start	Funding commitments, capex allocation and incremental design/qualification cost evidence.
Phase 1 2028-2030 ramp	Actual qualification, underutilization, break-even and full-scale conversion validate the existing operating reference.
Phase 2 readiness and first acceptance	Tooling, good-unit yield, customer acceptance and new net pricing/volume evidence distinguish available area from recognized services.

The next useful update should test these specific operating conditions. A broad AI spending headline alone does not establish another Amkor revenue pool; a distinct accepted-program or qualified-capacity bridge can support a new probability-weighted revision.

Source coverage and the evidence boundary

Financial statements and issuer releases are primary evidence. Management transcript reproductions are attributed statements with their precise completeness limits. The images are preserved source-page excerpts, not reconstructed exhibits. Whole-number report formatting does not alter original source decimals. Exact models and source hashes are retained locally.

Coverage item	Completed review / limitation
Latest filing and release	Latest Q2 financial anchors, cost/segment notes, shares, debt, contract liabilities, capex, working capital, relevant MD&A and risk sections read. This is not a claim that every legal or derivative detail was reviewed.
Two latest earnings calls	Q1 official full transcript read. Q2 full saved management reproduction read through closing; official PDF/audio verification unavailable.
Investor Day	All 520 SEC text lines, representing 71 slides, preserved and read. Original large slide PDF/images and full event Q&A/audio not available.
September conference	Visible Q&A through closing read; official calendar verifies event date. Transcription errors and a provider completeness caveat remain.
Original visuals	Original 2026 filing passages and dated 2018/2020 technical pages rendered directly from retained PDFs. No fabricated current-campus slide or yield figure.
Unknown operating inputs	Phase 2 package units, product mix, ASP, qualified yield, detailed tools, customer acceptance schedule, exact cost and financing terms remain unresolved.
Excluded future evidence	26 October Q3 actuals/call and any later publication.

The forward model separates official dates, attributed management direction and LOW estimates. Direct source identities and document scopes follow.

Direct source register

Source	Date / period	Document	Provenance and scope
A01	27 July 2026	Issuer Q2 release; actuals and Q3 outlook	Primary HTML original retained
A01P	27 July 2026	Saved Q2 release PDF	Preserved local original; remote byte identity not reverified
A02P	28 July 2026	Q2 Form 10-Q; quarter ended 30 June	Saved 63-page filing PDF; text checked against issuer/SEC; original decimals retained
A04	27 April 2026	Q1 earnings call	Official issuer-hosted LSEG PDF, 14 pages; full prepared/Q&A/closing read
A09	FY2025 filing	10-K business, Competition and asset lives	Primary HTML; complete Competition, PPE accounting policy and Note 8; selected business/technology review
A10	8 September 2026	Arizona Phase 2 announcement	Primary issuer HTML; area, investment and construction timing
A11	23 July 2026	NVIDIA strategic partnership	Primary issuer HTML; customer alignment and expected advance
A12	16 June 2026	TSMC long-term partnership	Primary issuer HTML; ten-year relationship, no global exclusive volume award
A13	Current site at retrieval	SWIFT product page	Primary corporate technical page, not a financial forecast
A19	September 2018	SWIFT data sheet	Original official PDF, 2 pages; historical technology illustration
A20	Summer 2020	Networking / advanced package article	Issuer-authored original PDF, 6 pages; historical technical mechanism
A18B	21 May 2026	Investor Day SEC text representation	All 520 lines preserved; original slide PDF/image bytes unavailable
A22B	9 September 2026	Management conference reproduction	Attributed CFO/management statements; visible closing with provider caveat
A23	27 July 2026	Q2 earnings-call reproduction	Saved full management reproduction through 39:33; official PDF/audio gap

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Method and artifact controls: current Framework 8.0; retained 20x forward GAAP OP; single common-share convention; calendar quarters; one active operating case with explicit timing states; full eighteen-quarter operating path and fifteen four-quarter valuation chains. The self-contained PDF and editable report embed the same original evidence and the same frozen model charts.